Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use $v=f\lambda$ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your model or result. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A system absorbs 70 J heat and does 30 J work.
Find its internal-energy change using $\Delta U=Q-W_{by}$.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

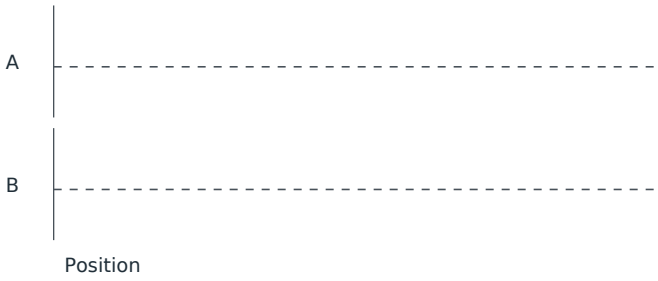
02 Worked example: explain the model

Reference: $v = f\lambda$; $\lambda = v/f$.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength 3 m and frequency 2 Hz travels at $v=f\lambda=6$ m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

Your turn: Draw both waves over the same 6 m length and label their frequencies.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

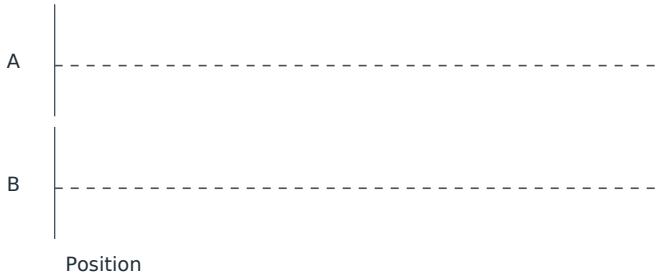
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two waves travel in the same ideal medium at the same speed. One has twice the frequency.
Compare wavelengths.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

04 Apply the physics

A wave simulator shows frequency 4 Hz and wavelength 2 m.
Calculate wave speed. Set the frequency to 8 Hz while keeping speed fixed.
Find the new wavelength and sketch both patterns over the same distance.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

A student writes v=f/λ.
Check units and explain why that is not a speed.

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: 1 2 3

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Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Sunlight carries energy across 4 million km of nearly empty space in a simplified example.
Identify the transfer mechanism and explain why matter is unnecessary.
Label temperatures, samples and relevant energy transfers.

Sentence starter: My claim is … . The diagram/data show …, so … .

Sample A

Sample B

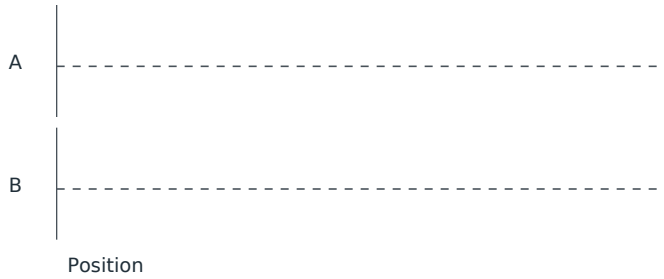
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

f=3 Hz,λ=5 m.
Find v.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Create three frequency-wavelength pairs that all give 12 m/s.
Sketch one pair.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3